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	The research project is aimed at studying thermoacoustic and buoyancy-driven transport in gases and in fluids near their critical points at normal and reduced gravity environments. It is envisioned that supercritical fluid reactors will play a vital role in the development of 'water recovery' and waste treatment systems in long duration manned space missions. The present four-year project (start date: December 8, 2003) combines computational and experimental studies of thermoacoustic waves and thermoacoustic convection in gases and in near-critical fluids. In the first two years of the project, significant advances have been made in both the model development and the experimental design and facility development. Large-scale detailed computational models were developed and tested to simulate, and understand the thermoacoustically driven flows and transport in gases and the generation and propagation of thermoacoustic waves in supercritical carbon dioxide. Both normal-gravity and reduced-gravity experiments are planned to validate the key		

Task Description:	features of the simulations. For normal-gravity experiments in gases and in supercritical fluids, two experimental facilities have been designed and built at Drexel University. We extended our computational fluid dynamic models (for predicting thermoacoustic waves in supercritical fluids) by incorporating the NIST Reference Database 12 for supercritical carbon dioxide properties. In the case of low-heat-diffusing, near-critical fluids, it has been recently shown that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. The piston-effect has been successfully captured in our simulations using the models developed. Normal gravity experiments were conducted for the generation and characterization of thermoacoustic waves in gases. Preliminary data agree well with model predictions. An experimental setup to generate and characterize the thermoacoustic waves in supercritical carbon dioxide has been designed and fabricated. Studies will be carried out under both normal and reduced gravity environments. Regarding reduced-gravity experiments at the NASA Glenn 2.2 s drop tower, we visited the facilities in August 2005. Modified experimental setups are now being designed that will satisfy the drop tower capabilities and guidelines. The present research is aimed at developing advanced computational tools to analyze interactions between thermoacoustic waves and viscous fluids in several problems of significant relevance to microgravity conditions. The specific objectives of the proposed research are the following: • Improve and refine a high-order computational model for the simulation of flow and heat transfer in gases and near-critical fluids induced by thermoacoustic waves • Perform direct numerical simulation of thermoacoustically driven gas flows in 2-D and 3-D enclosures • Completely characterize the behavior of the thermoacoustic waves (generation, propagation, interaction with solid walls and eventual dissipation) in gases and nar-critical fluids • Perform normal-gravity experiments for thermoacoustic waves in gases for model validation • Perform reduced-gravity experiments (short-time) in drop-towers or KC 135 aircraft for isolating buoyancy effects from thermoacoustic wave generated flows in gases • Select appropriate equation of state for near-critical fluids and incorporate the equation in the numerical model for flow simulation in such fluids • Perform direct numerical simulation of the piston effect in near-critical fluids in enclosures • Execute model for predicting the 'thermoacoustic' effects in near-critical fluids with and without gravitational effects • Perform normal-gravity experiments with near-critical CO2 in carefully designed experiments (which reduce gravitational effects) • Develop guidelines for the use of the mathematical models as a design and optimization tool for property and transport characterization in supercritical fluids and supercritical reactors. Successful completion of the research tasks will provide us important information regarding the design and operation of supercritical reactors. Such reactors are envisioned for waste management and water recovery in long duration manned space missions. The effects of thermoacoustic waves on transport and mixing in microgravity environments in such reactors willbe significant. The present research will be useful for the design of supercritical carbon dioxide reactors used in the food and pharmaceutical industries and for supercritical water reactors for waste management. The presence of gravitational field leads to both practical and fundamental limits of the resolution in critical phenomena experiments in fluids near the gas-liquid critical point. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. The resulting inhomogeneous distribution of the local properties of the fluid strongly affects the interpretation of critical phenomena experiments in fluids. Near the critical point, fluids exhibit anomalies in the transport of heat such that convection and buoyancy phenomena appear for even minute temperature gradients. In the case of low-heat-diffusing, near-critical fluids, it has been shown recently that thermoacoustic waves, viz. the 'piston effect' is responsible for fast heat transport. A significant question regarding the 'piston effect' is whether the fast heat transport is due to the strong buoyant convection induced by normal gravity in near-critical fluids or if it is due to the high compressibility of such fluids. Microgravity environment is thus ideal for studying such processes for fluids near the critical point where new fluid physics can be discovered.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Supercritical fluids exhibit a number of specific properties like large density, low viscosity, low thermal diffusivity. In addition, their isothermal compressibility and thermal expansion can become very large, especially when they approach the critical point. The highly variable properties of near-critical fluids make them very attractive for studying many interesting phenomena that hold for all fluids because of the critical universality. Fluids in their supercritical state are increasingly used by the food and waste management industry for their solubilization properties (e.g. supercritical carbon dioxide), as host of "cold" combustion (e.g. supercritical water), in energetics (supercritical thermal or nuclear power plants) and in astronautics (e.g. storage of cryogenic fluids). It is particularly challenging to understand the behavior, flow and transport in a fluid that is close to its critical point. Near the gas-liquid critical point, gravity induces a density gradient in the fluid. The resulting inhomogeneous distribution of the local properties of the fluid strongly affects the interpretation of critical phenomena experiments in fluids. The research results will provide valuable insights to the trasnport processes in supercritical reactors - whether they are used on eart or long duration manned space missions.

Task Progress:

We made significant progress in model development (for characterizing thermally generated acoustic waves in gases and near-critical fluids and in the design and fabrication of two experimental set ups to study thermally generated acoustic waves in gases and in supercritical CO2. We briefly describe the results obtained to date below:
Computational studies (a) Studies in ideal gas The effect of gravity on the generation and propagation of thermally generated acoustic wave induced flows in an enclosure was numerically investigated. The compressible unsteady Navier-Stokes equations were solved by a flux-corrected transport algorithm for the convection terms and by a central differencing scheme for the viscous and conduction terms. For the present simulations, the left wall of the enclosure is heated rapidly either in a spatially uniform or a non-uniform manner, while the right wall is held at the initial temperature of the gas. The horizontal walls of the enclosure are insulated. Under zero-gravity conditions, the spatially non-uniform heating is found to generate a vortical flow in the enclosure, similar to that found in buoyancy induced flows in a side-heated enclosure. For the conditions considered, the strength of the thermally generated acoustic wave induced flows is comparable to the buoyancy induced flows. Thermally generated acoustic wave induced flow fields were simulated using a 2D time-dependent compressible Navier-Stokes code. The results are discussed in a manuscript titled, “Flows Induced by Thermally Generated Acoustic Waves in an Enclosure: Effects of Gravity” by Y. Lin, B. Farouk and E. S. Oran (AIAA Journal of Thermophysics and Heat Transfer, in press). Our numerical model for ideal gas was successfully applied for predicting acoustic streaming in a resonator driven by a loudspeaker. The results will be presented in the 2005 ASME National Heat Transfer Conference, San Francisco, July 2005.

(b) Studies in supercritical carbon dioxide Under microgravity, heat transfer in supercritical fluids is expected to be very weak, due to the vanishing thermal diffusivity near the critical point. However, fast equilibration time was detected in past experiments because the bulk fluid is homogeneously heated by the thermally generated acoustic waves generated. This effect is known as the piston effect. The generation and propagation of a thermally generated acoustic wave in mildly supercritical carbon dioxide was investigated by solving the fully compressible form of the Navier-Stokes equations. Two infinite plates filled with supercritical carbon dioxide, where the right walls is thermally insulated, is considered as the computational domain. A thermally induced pressure wave is generated by heating the left wall. The wave repeatedly traverses the length between the left and right walls, and its amplitude eventually damps out due to the viscous and thermal losses within the fluid, thus heating the bulk fluid. Long-time results show thermal equilibration and wave damping in the enclosure, which validates the existence of the piston effect in supercritical carbon dioxide. Thermally generated acoustic wave generation and propagation in supercritical carbon dioxide was simulated using a one-dimensional compressible Navier-Stokes solver along with the NIST Reference Database 12 for the thermodynamic property relations. The results of the study are described in a manuscript discussed in a manuscript titled, “Generation and Propagation of Thermally generated acoustic waves in Supercritical Carbon Dioxide” by Z. Lei, B. Farouk, M. Aktas and E. S. Oran (submitted to the Physics of Fluids, May, 2005). We had earlier used a boundary condition based on wave theory derived by Poinsot and Lele (Poinsot and Lele 1992) to compute the density at walls. This boundary condition was found to predict weak thermally generated acoustic waves. Now we are using a different density boundary condition derived form the continuity equation (per our discussions with colleagues at the Naval Research Laboratory, Washington, DC). A comparison of the results using the Poinsot and Lele (PL) boundary condition with the continuity equation (CE) derived boundary condition follows. Comparing the results with the PL boundary condition with those with the CE boundary condition, it is evident that the overall shape and the characteristics of the thermally generated acoustic waves predicted by both boundary conditions are quite similar; however, subtle but important differences exist. The CE boundary condition predicts stronger pressure and temperature peaks for the thermally generated acoustic waves generated. Piston effect is an interesting mechanism of heat transfer in supercritical fluids. It is the result of propagation and damping of thermally generated acoustic waves in supercritical fluids. The comparison between the results show that the CE boundary condition produces stronger piston effect. We plan to use the CE (instead of the PL) boundary condition for density from now on in all of our simulations of thermally generated acoustic waves in gases and supercritical fluids. Some of our recent results on predicting thermally generated acoustic waves in supercritical CO2 were presented at the ASME National Heat Transfer Conference, July 2005 held at San Francisco. CA. The computed results will be compared to our planned measurements in a supercritical CO2 chamber (work in progress).

Experimental Studies (a) Studies in ideal gas In the existing experimental setup, we can accurately measure the pressure waves in the early time. The measured data shows the generation and propagation of thermally generated acoustic waves in the enclosure. aA new experimental chamber, has been designed and built. The following improvements are made: (1) all separated parts are connected by flanges and rubber gaskets; (2) the chamber is thermally insulated; (3) gas filling and venting connectors are added; (4) absolute pressure will also be measured. With the re-designed experimental setup, we plan to achieve the following goals: (1) eliminate the mass leakage and heat loss of the chamber; (2) run the experiments for different kinds of gas at different initial pressures. Following our trip to the NASA 2.2 s drop tower facilities at Cleveland, Ohio (August 7 – 9, 2005), and meetings with Mr. Eric Bauman, Manager 2.2 s drop tower facilities, and Ms. Nancy Hall, Research Scientist (and our program monitor), a new experimental setup for studying thermally generated acoustic waves in gases was designed. After the technical tour to the 2.2 s drop tower, we submitted a ‘support request’ to Mr. Bauman. According to the requirement of the Drop Tower, initial mechanical and electrical designs were made. To better understand the gravity effect on the thermally generated acoustic wave induced flow fields, we plan to add flow visualization experiments. Therefore, we are redesigning and improving our experimental setup. We have also requested data acquisition hardware (a Generic Power Distribution Module, GPDM and a Digital Design and Custom Solution, DDACS data logger) commonly used for data acquisition in drop tower tests from NASA. After the measurements with the GPDM and DDACS and flow visualization experiments are tested in the Hess Lab (Drexel University)first, micro-gravity experiments will be carried out in the 2.2 s drop tower shortly.

(b) Studies in supercritical carbon dioxide The CO2 will be pumped from a tank (800 psi) into the reactor to exceed the critical pressure (1100 psi) value. A heating tape will be used to increase the carbon dioxide temperature above the critical value (31.8ºC). The reactor (made of stainless steel) has also been fabricated. In one end of the reactor, we use a thin metal foil, which can be rapidly heated by a resistance-capacitance circuit, to produce thermally generated acoustic waves. We shall measure the instantaneous temperature by a fine-wire thermocouple and the pressure by a Bruel and Kjaer microphone. For safety, we employed several rupture discs on charged lines and on the test chamber. The rupture pressure is set to be 1500psi for all the rupture discs, which will allow us to do the experiments in pressure range of 1100 - 1500. We also designed and built a polycarbonate cage as shown to house the test chamber. For electrodes, thermocouples and the cable for the microphone, we use seals (connectors with glands) to prevent leakage. We used a machine thread for the end caps of reactor, and we used Teflon gaskets to seal the end cap. A special housing needs to be constructed for mounting the microphone to the test chamber. As we were choosing the pressure sensor, we met a few challenges: microphones are rated for atmospheric pressure, and have only been tested up to 160 psi, not 1100 psi (the critical pressure of CO2). Hydrophones, which measure acoustic waves in high pressure fluid environments (such as sonar), are too bulky for the present setup. We need to pressurize all components or isolate the pre-amplifier of the

	microphone from the high pressure zone, if we use a microphone to measure the pressure waves in supercritical carbon dioxide. We plan to use an adaptor to isolate the preamplifier from the high pressure in the chamber. Calibration of the pressure (acoustic) sensor at supercritical conditions will be done with data obtained from numerical simulations.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Lin, Y.; Farouk, B.' and Oran, E. S. "Thermally Generated Acoustic Wave Induced Flows in an Enclosure: Effects of Gravity" AIAA Journal of Thermophysics and Heat Transfer , Jan-2006
Articles in Peer-reviewed Journals	Aktas, M. K.; Farouk, B.; and Lin, Y. "Heat Transfer Enhancement by Acoustic Streamin in an Enclosure" ASME Journal of Heat Transfer , Jan-2006
Papers from Meeting Proceedings	Lei, Z.; Farouk, B.; and Oran, E. S. "Numerical Simulation of Piston Effect in Supercritical Carbon Dioxide" ASME National Heat Transfer Conference, July 2005, San Francisco, CA. Jul-2005